

Wartime tech innovations

Check out this Quora thread for a discussion on technology innovations that came about due to war on terror. <http://dgit.in/YtM7Kj>

Algorithms in Computer Science

Want to know which computer algorithms are the most ingenious. Check out this trending Quora thread: <http://dgit.in/Z8c4i3>

Overclocking your PC

Siddhant Sharma

readersletters@thinkdigit.com

Scared of overclocking your PC or don't know how to do it? Then don't worry as we'll walk you down this narrow and seemingly dangerous path via this workshop.

This OC'ing guide is not platform specific and is meant to be generic to any CPU (be it Intel or AMD) as everyone has different computer components from various part manufacturers. This guide will provide you an idea about OC'ing.

Important terminologies

Clocks:

Intel CPU's use something called as *Base Clock* and AMD CPU's use *Front Side Bus*, but both of them are very similar. We'll refer to this as BCLK. Your CPU speed is a multiple of this so, for example, If a CPU runs at a 22x multiplier then that essentially means 22x BCLK. So if your CPU has a 133 mHz base clock then your CPU speed is 22x133= 2.9GHz. For RAM it goes like: 10x133=1333mHz. AMD's Front Side Bus or FSB works in the same way but instead of having multipliers, they

contravene the drop in Vcore once load is applied to the CPU. Intel calls this drop "Vdroop". When running an Intel CPU with Turbo Boost, Vdroop can change or shift it's multiplier automatically based on how many cores are needed. If only two cores out of four are required, the CPU will run at a higher frequency. This can also require a higher voltage, so with LLC off (Vdroop on) the voltage will naturally be a bit higher because there is a smaller load as two cores require lower power than four. However, during overclocking, it is better to disable Turbo Boost as when this is done, enabling LLC can be beneficial for getting a stable overclock. But this isn't necessary if you're not planning to cross 4GHz.

You can decrease the multiplier and keep it very low on most CPUs these days but their highest limit depends on the CPU itself. Intel as well as AMD, both have a series of "unlocked" CPUs. Intel has the "K" series moniker to signify an unlocked CPU and AMD on the other hand has the "Black Edition" series. Unlocked CPU's basically provide an ability to raise the multiplier to a certain high level which makes overclocking that CPU easy as we don't need to change the base clock or interfere with the RAM or QPI, if that's the case.

Note: Remember that the IMC/VTT/QPI voltage must be kept within 0.5V of DRAM voltage. DRAM Voltage is directly related to your RAM. RAM should always be set to the recommended manufacturer speeds. If for example, you might have bought a 2x 2GB 1600 MHz CL9 1.5V RAM kit. The 1.5V symbolises the DRAM voltage needed for that speed of 1600 MHz and timings of CL9.

Overclocking

Now starting up, enter your BIOS >Load defaults > Save and Exit. Then when the POST screen shows up, enter the BIOS again. Change or set your RAM settings to match the manufacturer specifications. This is easy as you just need to go to your



Disclaimer: Overclocking can seriously damage your hardware or burn your components. Overclocking depreciates the quality of hardware significantly, depending on how far you've pushed your components. If you do proceed to overclock, the life-expectancy of your components will be lowered. It may also VOID the warranty of some of your components especially on branded computers. Digit or 9.9 media and its employees will not be held liable for any damage that occurs, directly or indirectly, because of this article.

It will not only cover CPU OC'ing but also include pointers for tweaking your RAM.

Components required

Before you start OC'ing, there are a couple of things that you will absolutely need. The first is a motherboard that supports OC. The second most important thing you need is a good CPU which can be OC'ed. OC'ing heats up components so you need an aftermarket air or water cooler or even LN2 (Liquid Nitrogen) based cooling for extreme OC'ing. To know more about components required for overclocking, check out this month's Agent 001.

Utilities required

You can't just overclock your PC and start using it right away without checking its stability. We use Prime95 and OCCT for stress testing and monitoring programs used by us are CPU-Z and Real Temp. You can find a collection of almost 40+ overclocking and monitoring software and utilities in this month's DVD.

have ratios. When we say here "change the multiplier" consider this to also mean the ratio if you have an AMD rig.

Voltages:

Your CPU needs stable power and the settings related to increasing or decreasing the amount of voltage to be supplied to the CPU resides under the Vcore label (core voltage). Keep in mind that all CPU manufacturers provide specification sheets that mention the highest Vcore a CPU can handle, so you need to make sure to check that for your particular CPU before touching the Vcore settings. Intel also has voltage settings for IMCs (Integrated Memory Controller), sometimes VTT and also QPI (QuickPath Interconnect). AMD on the other hand has NB voltage (North-bridge Voltage) which is similar to IMC and also to QPI. Increasing the NB voltage along with the Vcore helps keep stability.

Others

Load Line Calibration (or LLC) in the BIOS is how the motherboard will try to